

**A REPORT
ON
INTEGRATED MACHINE LEARNING
FOR
REGRESSION AND CLASSIFICATION**

Submitted by

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in partial fulfillment for the award of the

degree of

**BACHELOR OF TECHNOLOGY
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CERTIFICATE

This is to certify that the report of **CSE7000 – Internship** on
**“INTEGRATED MACHINE LEARNING FOR REGRESSION
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I hereby declare that the work, which is being presented in the report entitled **“INTEGRATED MACHINE LEARNING FOR REGRESSION AND CLASSIFICATION”** in partial fulfillment for the award of Degree of **Bachelor of Technology in Computer Science and Engineering**, is a record of my own investigations carried under the guidance of **Mr.Sakthi S**, Assistant Professor and **Ms.Shaik Salma Begum**, Assistant Professor, Presidency School of Computer Science and Engineering, Presidency University, Bengaluru.

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INTERNSHIP COMPLETION CERTIFICATE



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ABSTRACT

The Machine Learning Internship at Internspark provided an opportunity to gain practical knowledge by working with real datasets and implementing Machine Learning techniques. As part of the internship, three projects were carried out: House Price Prediction, Titanic Survival Prediction, and Iris Flower Classification. Python was used for the implementation, along with Pandas, NumPy, Matplotlib, and Scikit-learn for data processing, analysis, visualization, and model development. Through these projects, the practical steps involved in developing Machine Learning solutions were understood and applied..

The Titanic Survival Prediction project aimed to analyze passenger information and predict survival outcomes. The dataset contained details such as age, gender, ticket class, fare amount, and family-related information. These attributes were examined carefully to identify useful features for prediction. Data preparation steps, including handling missing values and selecting important variables, were performed before developing the model.

The prepared data was separated into training and testing sets, and a classification algorithm was applied to build the prediction model. The model learned the relationship between passenger characteristics and survival status from the training data. Once trained, it was used to classify passengers as survived or not survived. The prediction results were analyzed using classification measures to understand the capability of the model in making accurate decisions.

The Iris Flower Classification project focused on identifying flower species from their physical measurements. The dataset consisted of four

important characteristics: sepal length, sepal width, petal length, and petal width, which belonged to three different Iris species. The dataset was examined to understand the distribution of these measurements and the differences among the species. Visualization methods were used to observe how each feature contributed to distinguishing the flowers.

Following the analysis stage, the dataset was divided into training and testing data and used to build a classification model. The model learned the patterns associated with each species and applied this knowledge to classify new flower samples. The predicted results were evaluated to determine the model's accuracy and its ability to correctly identify the species based on the given measurements.

The House Price Prediction project was carried out to estimate the selling price of houses using different property-related features available in the dataset. Initially, the dataset was studied to understand the available information and identify the factors influencing house prices. Data cleaning and preparation techniques were applied to make the dataset suitable for Machine Learning tasks. Graphical analysis was performed to examine patterns, trends, and relationships among various features, which helped in understanding the characteristics of the data.

After completing the preparation stage, the dataset was divided into training and testing data. A regression model was developed using the training data to learn the relationship between house features and their prices. The trained model was then used to estimate house prices for unseen data. The predicted outputs were compared with actual values, and suitable evaluation methods were used to measure the effectiveness and reliability of the model.

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CHAPTER 1

INTRODUCTION AND COMPANY PROFILE

1.1 Industry Background

Machine Learning is a major branch of Artificial Intelligence that enables computer systems to learn from data, identify patterns, and make predictions without being explicitly programmed for every task. It is widely used in industries such as finance, healthcare, e-commerce, transportation, education, and business analytics. The increasing use of data and automated decision-making has made Machine Learning an important technology in modern industries. The internship provided practical exposure to Machine Learning concepts and their application to real-world datasets.

1.2 Organizational Hierarchy

InternSpark follows a structured internship environment where mentors and coordinators guide interns throughout their assigned activities. The mentors provide technical guidance, project instructions, and feedback, while interns are responsible for implementing the assigned tasks and completing the projects. This structure supports effective communication, continuous learning, and successful completion of internship objectives.

1.3 Department Role

The Machine Learning department focuses on applying data-driven techniques to develop prediction and classification solutions. The work involves understanding datasets, preprocessing data, performing exploratory analysis, selecting suitable algorithms, training models, generating predictions, and evaluating model performance. During the internship, these activities were carried out through practical Machine Learning projects.

1.4 Problem Statement

The internship focused on solving different prediction and classification problems using Machine Learning techniques. The main problem was to process datasets, identify relevant patterns and features, and develop models capable of producing accurate predictions. The projects involved house price prediction, Titanic survival prediction, and Iris flower classification.

1.5 Objectives

The main objective of the internship was to gain practical knowledge of Machine Learning and understand the complete process of developing prediction and classification models. The internship aimed to develop skills in data preprocessing, visualization, algorithm selection, model training, prediction, and performance evaluation while improving programming, analytical thinking, and problem-solving abilities.

1.6 Scope Boundaries

The scope of the internship was limited to developing and evaluating the three selected Machine Learning projects using the available datasets and tools. The work focused on data preprocessing, model development, prediction, and performance evaluation using Python and Jupyter Notebook. Advanced production deployment, real-time systems, and large-scale enterprise implementation were outside the scope of the internship.

CHAPTER 2

REQUIREMENT ANALYSIS AND SYSTEM DESIGN

2.1 Software Requirements Specification (SRS)

The Software Requirements Specification defines the basic requirements needed to develop and execute the Machine Learning projects completed during the internship. The projects required datasets containing relevant features for prediction and classification tasks, along with suitable software tools for data preprocessing, visualization, model development, and evaluation. Python was used as the primary programming language, while Jupyter Notebook provided an interactive environment for writing and executing the programs. The required Python libraries included Pandas and NumPy for data handling and numerical operations, Matplotlib for data visualization, and Scikit-learn for implementing and evaluating Machine Learning models.

2.2 Functional Requirements

The functional requirements describe the operations that the Machine Learning system should perform for the three projects. The system should be able to load the required datasets and display the available information for analysis. It should perform data preprocessing, including handling missing values, preparing features, and converting data into a suitable format for model training. The system should allow suitable Machine Learning models to be trained using the prepared datasets and should generate predictions for previously unseen data. Finally, the system should evaluate the trained models using appropriate performance measures and display the obtained results for analysis.

For the Titanic Survival Prediction and Iris Flower Classification projects, the system should perform classification and predict the appropriate category for the given input. For the House Price Prediction project, the system should perform regression and generate a numerical house price based on the available input features.

2.3 Non-Functional Requirements

The non-functional requirements define the expected quality and usability of the Machine Learning implementation. The system should provide reliable results when the datasets are properly prepared and the selected models are trained correctly. The implementation should be understandable, maintainable, and organized into logical stages such as data loading, preprocessing, training, prediction, and evaluation. The system should also provide results within a reasonable execution time and should be capable of handling the datasets used in the projects. The implementation should be reproducible so that the same data and configuration can produce consistent results.

2.4 Hardware and Software Requirements

As shown in Table 2.1 the Machine Learning projects were developed using a computer system capable of running Python and Jupyter Notebook. A standard computer with sufficient processing capability, RAM, and storage was used for dataset processing and model training. The software environment consisted of Python and Jupyter Notebook along with the required Machine Learning and data analysis libraries. Pandas and NumPy were used for data manipulation and numerical calculations, Matplotlib was used to visualize the data and results, and Scikit-learn was used for preprocessing, model training, prediction, and evaluation.

Component	Choice	Purpose
Programming Language	Python	Used for implementing the machine learning algorithms and project logic.
Development Environment	Jupyter Notebook	Used for writing, executing, and documenting Python code interactively.
Data Processing	Pandas	Used for loading, cleaning, manipulating, and analyzing datasets.
Numerical Computing	NumPy	Used for numerical operations and handling arrays and mathematical computations.
Data Visualization	Matplotlib	Used to create graphs and visualizations for data analysis.
Machine Learning Library	Scikit-learn	Used for data preprocessing, model training, prediction, and performance evaluation.

Table 2.1 Hardware and software stack purpose

2.5 System Architecture and Workflow

As shown in Figure 2.1 the overall architecture of the Machine Learning projects follows a sequential workflow. The process begins with collecting or loading the required dataset, followed by understanding the structure and characteristics of the data. The dataset is then preprocessed by handling missing values, preparing features, and converting the data into a suitable format.

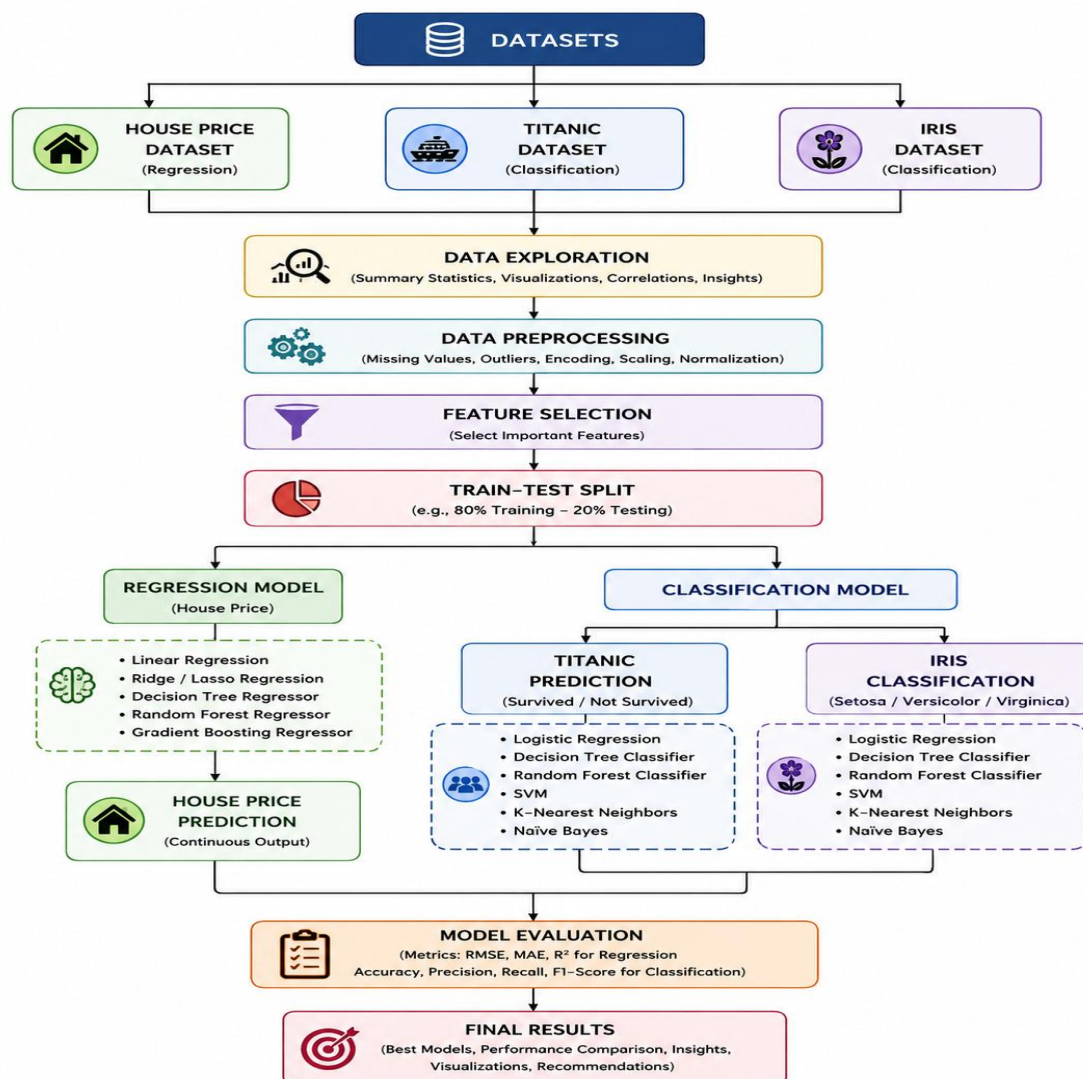


Fig 2.1 System Architecture

The prepared data is divided into appropriate training and testing sets, after which a suitable Machine Learning model is selected and trained. The trained model is then used to make predictions on the test data. Finally, the predictions are compared with the actual values and evaluated using suitable performance metrics.

The general workflow can be represented as:

Dataset → Data Exploration → Data Preprocessing → Feature Selection → Train-Test Split → Model Training → Prediction → Model Evaluation → Results

This workflow was applied according to the requirements of each project. Classification techniques were used for Titanic Survival Prediction and Iris Flower Classification, while regression techniques were used for House Price Prediction.

2.6 Data Flow Schema

As shown in Figure 2.2, the data flow begins with dataset input and proceeds through data preprocessing, feature selection, model training and prediction, and finally model evaluation to obtain the final results.

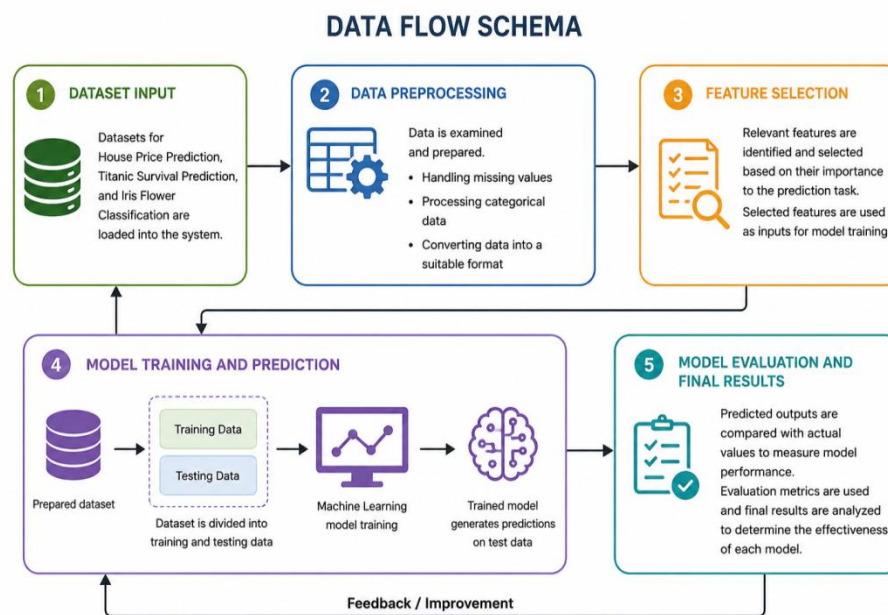


Fig 2.2 Data Flow Diagram

1. Dataset Input

The required datasets for House Price Prediction, Titanic Survival Prediction, and Iris Flower Classification are loaded into the system. The datasets contain the features and target values required for developing the respective Machine Learning models.

2. Data Preprocessing

The collected data is examined and prepared for model development. This step includes handling missing values, processing categorical data where required, and converting the dataset into a suitable format.

3. Feature Selection

Relevant features are identified and selected based on their importance to the prediction task. The selected features are used as inputs for training the Machine Learning models.

4. Model Training and Prediction

Prepared datasets are divided into training and testing data, and suitable Machine Learning models are trained. The trained models are then used to generate predictions for the test data.

5. Model Evaluation and Final Results

The predicted outputs are compared with the actual values to measure model performance. Suitable evaluation metrics are used, and the final results are analyzed to determine the effectiveness of each model.

CHAPTER 3

TECHNICAL IMPLEMENTATION AND CORE MODULES

3.1 Step-by-Step Algorithms

The algorithms used in the machine learning projects were implemented through a systematic sequence of steps. The process begins with loading the dataset, followed by data exploration and preprocessing. Relevant features are selected, and the dataset is divided into training and testing sets. The selected machine learning model is then trained using the training data, predictions are generated using the test data, and the model performance is evaluated using appropriate metrics.

3.2 Code Design pattern

The code was developed using Python in Jupyter Notebook and organized into separate sections for each stage of the machine learning workflow. Libraries were imported first, followed by dataset loading, data preprocessing, model creation, training, prediction, and evaluation. This structured approach made the code easier to understand, execute, debug, and modify.

The data flow begins with importing the required Python libraries for data analysis, preprocessing, visualization, and machine learning. The dataset is then loaded into the Jupyter Notebook and examined to understand its structure and contents. Data preprocessing is performed by handling missing values, removing unnecessary data, and preparing the dataset for further analysis.

Relevant features are selected based on their importance to the prediction task. A suitable machine learning model is then created and trained using the prepared data. After training, the model makes predictions on the test data. The performance of the model is evaluated using suitable evaluation metrics, and the final results are analyzed to determine the effectiveness and accuracy of the developed machine learning model.

3.3 Database Schema

The datasets were organized in a tabular structure consisting of rows and columns. Each row represented an individual observation, while the columns represented different features and the target variable. The data structure was examined and prepared before applying machine learning algorithms. Since the implementation was carried out using datasets in Jupyter Notebook, a separate database schema was not used.

3.4 Workflow Logic

The workflow followed a sequential machine learning process from raw data to the final result. The major stages were data loading, exploration, preprocessing, feature selection, train-test splitting, model training, prediction, and evaluation. This workflow ensured that the data was properly processed before being provided to the machine learning model.

3.5 Core Module Implementation

The project was divided into core modules based on the different stages of machine learning. The data processing module handled cleaning and preparation, the model module handled training and prediction, and the evaluation module measured model performance. These modules worked together to produce the final output.

3.6 Model Evaluation and Performance Analysis

The trained models were evaluated using appropriate performance metrics. Classification models were assessed using Accuracy, Precision, Recall, and F1-score, while regression models were evaluated using MAE, MSE, and R^2 score. The evaluation results were used to understand the effectiveness of the implemented models and compare their performance.

CHAPTER 4

TESTING ,VERIFICATION AND QUALITY ASSURANCE

4.1 Testing and Verification

As shown in Table 4.1, the testing and verification process ensures that the datasets and machine learning models produce reliable and accurate results

Test Category	Sample Query	Expected Behaviour
Dataset Loading	Is the dataset loaded correctly?	Dataset should load successfully without errors.
Data Preprocessing	Are missing values and incorrect data handled?	Data should be cleaned and prepared properly.
Feature Selection	Are the relevant features selected?	Required features should be selected correctly.
Model Training	Is the Machine Learning model trained successfully?	Model should train without errors.
Model Prediction	Does the model predict correctly on test data?	Predictions should be generated accurately.

Table 4.1 Testing and Verification

Testing was performed to verify that the machine learning models produced reliable and consistent results. The datasets were checked after preprocessing to ensure that missing values, incorrect data types, and unwanted records were handled appropriately. The trained models were then tested using unseen test data to verify their prediction performance.

4.2 Model Validation

The machine learning models were validated using separate training and testing datasets. The training data was used to develop the models, while the testing data was used to evaluate their performance on unseen observations. This helped determine whether the models were able to generalize effectively beyond the training data.

4.3 Performance Metrics

Appropriate evaluation metrics were used to measure the performance of the developed models. For classification tasks, Accuracy, Precision, Recall, and F1-score were considered. For regression tasks, Mean Absolute Error (MAE), Mean Squared Error (MSE), and R^2 score were used to measure prediction accuracy and error.

4.4 Model Comparison

When multiple machine learning algorithms were tested, their evaluation results were compared to determine the most suitable model. The comparison was based on the calculated performance metrics, and the model providing better results for the respective problem was selected.

4.5 Error Verification and Quality Assurance

The implementation was checked for errors during data preprocessing, model training, prediction, and evaluation. The generated outputs and evaluation metrics were verified to ensure that the code executed correctly and produced meaningful results. This process helped improve the reliability and quality of the machine learning implementation.

4.6 Testing Results

The final testing results were obtained by applying the trained models to the test datasets. The predicted outputs were compared with the actual values using the selected evaluation metrics. The results demonstrated the effectiveness of the implemented machine learning models and helped identify areas where model performance could be improved.

CHAPTER 5

RESULTS ,OUTPUTS ARTIFACTS AND DISCUSSION

5.1 Output Artifacts

The output artifacts generated during the implementation of the three Machine Learning projects are presented in this chapter. These include prediction outputs, classification results, graphs, evaluation results, and other visualizations generated using Jupyter Notebook. The artifacts demonstrate the practical implementation and final results obtained from each project.

5.2 Titanic Survival Prediction Results

The Titanic Survival Prediction model generated survival predictions based on passenger-related features such as age, gender, passenger class, and fare. The prediction results demonstrate the ability of the trained classification model to distinguish between passengers who survived and those who did not. The generated outputs and evaluation results are presented to analyze the effectiveness of the model.

5.3 House Price Prediction Results

The House Price Prediction model generated estimated house prices using property-related features. The model analyzed the given input variables and produced continuous numerical predictions. The prediction outputs and evaluation results demonstrate the application of regression techniques for estimating property prices.

5.4 Iris Flower Classification Results

The Iris Flower Classification model classified flowers into their respective species using sepal and petal measurements. The classification outputs demonstrate how the trained model identified different iris species based on their input features. Graphical representations of the classification results are included to provide a clear understanding of the model output.

5.5 Evaluation and Comparison

The performance of the developed models was analyzed using suitable evaluation metrics. Classification models were evaluated using Accuracy, Precision, Recall, and F1-score, while the regression model was evaluated using MAE, MSE, and R^2 score. The obtained results were compared to understand the effectiveness of each model and identify areas for improvement.

As shown in Figure 5.1, the graph compares the survival of male and female passengers. It shows that more females survived than males, while the number of males who did not survive was much higher.

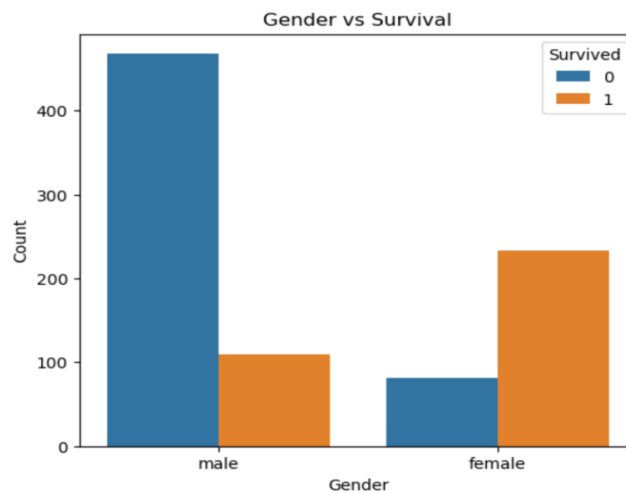


Fig 5.1 – Titanic Survival Graph

As shown in Figure 5.2, the graph represents the distribution of house prices in the dataset. Most houses fall within the lower-to-middle price range, while only a small number of houses have very high prices.

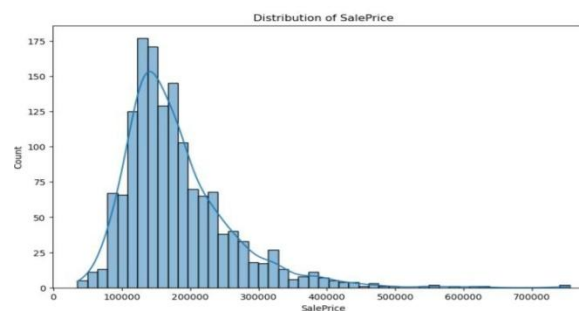


Fig 5.2 – House Price Prediction

As shown in Figure 5.3, the graph represents the relationships between different Iris flower features. It shows how the flower species can be separated based on their measurements, helping in the classification of different Iris flower species.

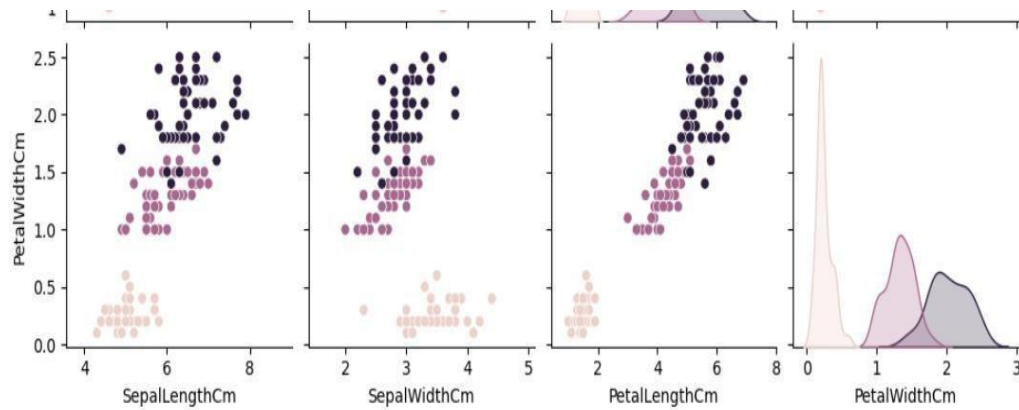


Fig 5.3 Iris Flower Classification Graph

5.6 Discussion of Results

The results obtained from the three projects demonstrate the practical application of Machine Learning to different types of prediction problems. Iris Flower Classification and Titanic Survival Prediction involved classification tasks, while House Price Prediction involved regression. The results show how appropriate preprocessing, feature selection, model training, and evaluation techniques contribute to achieving reliable predictions.

CHAPTER 6

CHALLENGES ENCOUNTERED AND SOLUTIONS APPLIED

6.1 Data Quality and Missing Values

One of the major challenges was handling datasets containing missing, inconsistent, or incomplete values. Missing values could negatively affect the training process and reduce prediction accuracy. This issue was addressed by identifying null values during data preprocessing and applying suitable techniques such as filling missing values with appropriate statistical values or removing unnecessary records.

6.2 Data Preprocessing

The datasets contained different types of features, including numerical and categorical data. Machine Learning algorithms require data in a suitable numerical format. Therefore, categorical features were converted into numerical representations using encoding techniques, while numerical features were processed appropriately. This ensured that the datasets were compatible with the selected Machine Learning algorithms.

6.3 Model Performance

Another challenge was selecting a suitable Machine Learning algorithm that could provide reliable predictions. Different models can produce different results depending on the characteristics of the dataset. To overcome this issue, multiple algorithms were trained and evaluated using appropriate performance metrics. The model with better evaluation results was selected for the final prediction.

6.4 Overfitting and Generalization

Overfitting was a potential challenge, particularly when models performed very well on training data but did not generalize effectively to unseen data. To reduce this problem, the datasets were divided into training and testing sets, and model performance was evaluated using unseen test data. Parameters such as the number of estimators and model complexity were also considered while training the models.

6.5 Computational and Training Issues

Training Machine Learning models can require considerable computational resources, especially when working with ensemble algorithms. Efficient preprocessing, suitable model parameters, and controlled model complexity were used to reduce unnecessary computation. This helped maintain a balance between model performance and training time.

6.6 Evaluation and Validation

Another challenge was determining whether the trained models were performing reliably. A single performance measure may not provide a complete understanding of a model's effectiveness. Therefore, suitable evaluation metrics were used depending on the project. Classification models were evaluated using metrics such as accuracy, precision, recall, and F1-score, while regression models were evaluated using MAE, MSE, and R^2 score.

6.7 Technical Remedies Adopted

The technical challenges encountered during the projects were addressed by applying suitable preprocessing, validation, and optimization techniques. Missing or inconsistent data was handled through data cleaning and appropriate preprocessing methods, while categorical data was converted into numerical form using encoding techniques. Feature scaling was applied where required to improve model performance, and different machine learning algorithms were trained and evaluated to select the most suitable model. Model performance was verified using appropriate evaluation metrics such as accuracy, precision, recall, F1-score, MAE, MSE, and R^2 .

CHAPTER 7

CONCLUSION AND FUTURE ENHANCEMENT

7.1 Conclusion

The Machine Learning internship provided valuable practical experience in applying machine learning concepts to real-world datasets. During the internship, three projects were successfully completed: Iris Flower Classification, Titanic Survival Prediction, and House Price Prediction. Each project focused on a different machine learning problem and helped develop a clear understanding of the machine learning development process.

The Iris Flower Classification project focused on predicting the species of an iris flower based on sepal and petal measurements. It provided practical experience in classification, data preprocessing, model training, prediction, and performance evaluation.

The Titanic Survival Prediction project focused on predicting whether a passenger survived the Titanic disaster using features such as passenger class, gender, age, and fare. This project provided experience in handling missing values, preprocessing categorical and numerical data, training classification models, and evaluating their performance using accuracy, precision, recall, and F1-score.

The House Price Prediction project focused on predicting house prices using property-related features. It provided practical experience with regression techniques, feature preparation, model training, prediction, and evaluation using metrics such as Mean Absolute Error (MAE), Mean Squared Error (MSE), and R^2 score.

Overall, these projects helped in understanding the complete machine learning workflow, including data preprocessing, exploratory data analysis, feature preparation, model selection, model training, prediction, and performance evaluation. The internship also strengthened practical knowledge of Python and libraries such as Pandas, NumPy, Matplotlib, and Scikit-learn.

7.2 Future Requirements

The three machine learning projects can be further enhanced to improve their accuracy, usability, scalability, and real-world applicability. The future enhancements can include model optimization, larger datasets, web application integration, Dockerization, CI/CD implementation, cloud deployment, and model monitoring.

These improvements can help transform the projects from basic machine learning implementations into more complete and industry-oriented applications. The enhanced systems can provide better predictions, improved user interaction, and easier deployment and maintenance.

7.3 Model Optimization

The machine learning models developed for the three projects can be further optimized to achieve better performance. Hyperparameter tuning techniques such as Grid Search and Randomized Search can be applied to identify suitable parameter combinations for each model.

Different algorithms can also be trained and compared using appropriate evaluation metrics. Feature selection and feature engineering can be performed to identify important features and improve the quality of the input data. These techniques can help increase prediction accuracy, reduce errors, and improve the models' ability to perform well on unseen data.

7.4 Larger and more Diverse Datasets

The projects can be further improved by using larger and more diverse datasets. A larger dataset can provide more information for the models to learn from and can improve their ability to generalize to new data.

For the Iris Flower Classification and Titanic Survival Prediction projects, additional records and relevant features can be included. For the House Price Prediction project, data from different locations, property types, sizes, and price ranges can be incorporated. Using more diverse data can make the models more reliable and suitable for real-world applications.

7.5 Industry skills Acquired

During the internship, practical skills were acquired in Python programming, data preprocessing, exploratory data analysis, data visualization, machine learning model development, training, and performance evaluation. The internship also improved problem-solving, debugging, documentation, and technical reporting skills through hands-on implementation of real-world machine learning projects.

7.6 Enterprise readiness

The internship provided an understanding of industry-oriented machine learning workflows, starting from data preparation and analysis to model development and evaluation. It also developed the ability to write structured and maintainable code, document technical work effectively, validate model performance, and understand modern deployment practices such as Dockerization, CI/CD, and cloud deployment.

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- [12] GitHub Repository – *Machine Learning Internship Projects*.
House Price Prediction: <https://github.com/SanikaHY/House-Price-Prediction>
Iris Flower Classification: <https://github.com/SanikaHY/iris-flower-classification>
Titanic Survival Prediction: <https://github.com/SanikaHY/Titanic-Survival-Prediction>

APPENDICES

APPENDIX A

SOURCE CODE

A.1 Titanic Survival Prediction

Sample code

```
# Create and train model
model = LogisticRegression(max_iter=1000)
model.fit(X_train, y_train)

# Prediction
y_pred = model.predict(X_test)

# Calculate metrics
accuracy = accuracy_score(y_test, y_pred)
precision = precision_score(y_test, y_pred)
recall = recall_score(y_test, y_pred)
f1 = f1_score(y_test, y_pred)

# Display results
print("=====")
print("      TITANIC MODEL RESULTS")
print("=====")
print("Accuracy  :", round(accuracy, 4))
print("Precision :", round(precision, 4))
print("Recall    :", round(recall, 4))
print("F1 Score  :", round(f1, 4))
print("=====")
```

Fig A.11 Source Code

As shown in Fig A.11 trains a Logistic Regression model on the Titanic training data and predicts whether passengers survived. It then evaluates the model using Accuracy, Precision, Recall, and F1 Score to measure its performance.

Sample Output

```
=====
              TITANIC MODEL RESULTS
=====
Accuracy   : 0.8101
Precision  : 0.7857
Recall     : 0.7432
F1 Score   : 0.7639
=====
```

Fig A.12 Sample Output

As shown in Figure A.12 trains a Logistic Regression model on the Titanic training

data and predicts whether passengers survived. It then evaluates the model

using Accuracy, Precision, Recall, and F1 Score to measure its performance.

A.2 Iris Flower classification

Sample code

	Model	Accuracy
0	Logistic Regression	0.966667
1	SVM	1.000000

```
best_model = results.loc[results['Accuracy'].idxmax()]

print("Best Model:", best_model['Model'])
print("Best Accuracy:", best_model['Accuracy'])

Best Model: SVM
Best Accuracy: 1.0

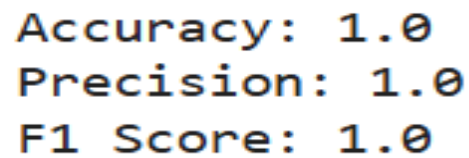
print("Logistic Regression:", accuracy_score(y_test, y_pred))
print("SVM:", accuracy_score(y_test, svm_pred))

Logistic Regression: 0.9666666666666667
SVM: 1.0
```

Fig A.21 Source Code

As shown in Fig A.22 tells the accuracy of Logistic Regression and SVM models was compared to determine their performance. The model with the highest accuracy was selected as the best-performing model

Sample Output



```
Accuracy: 1.0
Precision: 1.0
F1 Score: 1.0
```

Fig A.22 Sample Output

As shown in Fig A.22 ,describes about iris flower classification's accuracy,precision and F1 score

A.3 HousePricePrediction

Sample code

```
# Create model
model = RandomForestRegressor(
    n_estimators=100,
    random_state=42
)

# Train model
model.fit(X_train, y_train)

# Prediction
y_pred = model.predict(X_test)

# Evaluation
mae = mean_absolute_error(y_test, y_pred)
mse = mean_squared_error(y_test, y_pred)
r2 = r2_score(y_test, y_pred)
```

Fig A.31 Source Code

As shown in Fig A.31 the Random Forest Regressor was trained to predict house prices using the prepared training data.

The model was evaluated using MAE, MSE, and R^2 score to measure its prediction performance.

Sample Output

As shown in Fig A.32 the Random Forest Regression was trained to predict house prices using the prepared training data.

The model was evaluated using MAE, MSE, and R^2 score to measure its prediction performance.

```
Mean Absolute Error: 17890.921746575343
Mean Squared Error: 896173959.223158
R2 Score: 0.8831635141692619
```

Fig A.32 Sample Output

As shown in Fig A.32 the Random Forest Regression was trained to predict house prices using the prepared training data.

The model was evaluated using MAE, MSE, and R^2 score to measure its prediction performance.

APPENDIX B

USER HANDBOOK

B.1 Setting Up the Environment

B.1.1 Install Python on the system and open Jupyter Notebook

B.1.2 Create or open the required project notebook.

B.1.3 Install and import the required libraries such as Pandas, NumPy

B.1.4 Once the environment is ready, load the required dataset and execute the project.

B.2 Iris Flower Classification

B.2.1 Open the Iris Flower Classification notebook.

B.2.2 Import the required Python libraries such as Pandas, NumPy, Matplotlib, and Scikit-learn.

B.2.3 Load the Iris Flower dataset into the notebook.

B.2.4 Examine the dataset and check the available features and target values.

B.2.5 Perform the required data preprocessing and separate the input features from the target variable.

B.2.6 Split the dataset into training and testing data.

B.2.7 Train the selected classification model using the training data.

B.2.8 Provide the test data to the trained model to generate predictions.

B.2.9 Evaluate the model using suitable classification metrics such as accuracy.

B.2.10 Observe the predicted iris flower species and the model evaluation results.

B.3 Titanic Survival Prediction

B.3.1 Open the Titanic Survival Prediction notebook.

B.3.2 Import the required Python libraries.

B.3.3 Load the Titanic dataset into the notebook.

B.3.4 Examine the dataset and identify missing values and relevant passenger features.

B.3.5 Perform data preprocessing by handling missing values and converting categorical data into a suitable numerical format.

B.3.6 Select the required input features and survival status as the target variable.

B.3.7 Split the dataset into training and testing data.

B.3.8 Train the selected classification model using the training data.

B.3.9 Use the trained model to predict the survival status of passengers in the test data.

B.3.10 Evaluate the model using accuracy, precision, recall, and F1-score.

B.3.11 Observe the predicted survival results and the model performance.

B.4 House Price Prediction

B.4.1 Open the House Price Prediction notebook.

B.4.2 Import the required Python libraries.

B.4.3 Load the House Price dataset into the notebook.

B.4.4 Examine the dataset and identify the property-related features and target house price.

B.4.5 Perform the required data preprocessing and prepare the input features.

B.4.6 Separate the input features from the target variable.

B.4.7 Split the dataset into training and testing data.

B.4.8 Train the selected regression model using the training data.

B.4.9 Use the trained model to predict house prices for the test data.

B.4.10 Evaluate the model using Mean Absolute Error (MAE), Mean Squared Error (MSE), and R^2 score.

Appendix C

WEEKLY WORK LOGS

As shown in Table C. 1 the weekly work log presents the activities completed during the eight-week internship. It summarizes the major tasks and outcomes achieved throughout the Machine Learning projects.

Week	Work Completed	Outcome
Week 1	Learned Machine Learning fundamentals and set up Python and required libraries.	Environment successfully prepared.
Week 2	Explored the Iris dataset, performed preprocessing, and trained a classification model.	Iris classification model developed.
Week 3	Preprocessed the Titanic dataset and developed a survival prediction model.	Titanic survival model developed and evaluated.
Week 4	Prepared the House Price dataset and trained a regression model.	House price prediction model developed.
Week 5	Compared machine learning models using suitable evaluation metrics.	Model performance analyzed and compared.
Week 6	Created visualizations and analyzed patterns in the datasets and results.	Data patterns and results interpreted effectively.
Week 7	Tested all three projects, fixed errors, and verified the outputs.	Models and outputs successfully verified.
Week 8	Prepared documentation and finalized the three projects and internship report.	Projects and report successfully completed.

Table C.1 Weekly Work Log

